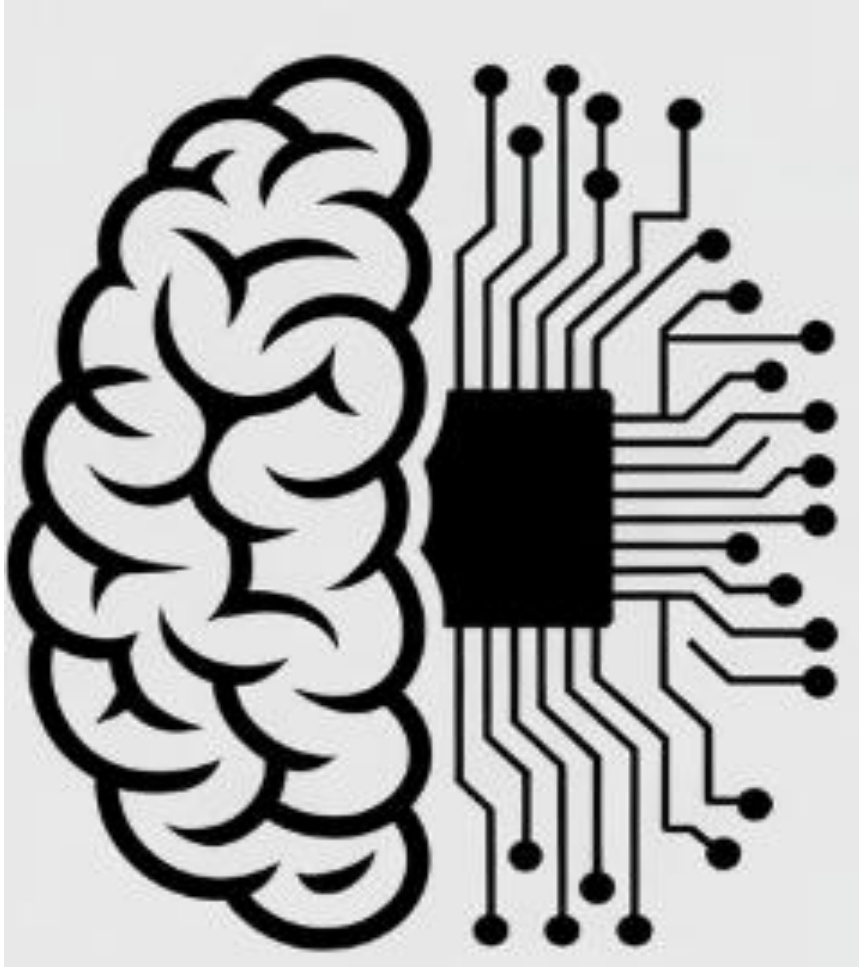
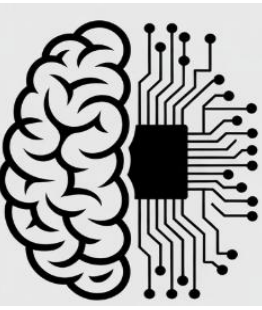




Artificial Intelligence

Dr. Sobia Tariq Javed



Administrative Information

- Course Instructor
 - Dr. Sobia Tariq Javed
- Email-ID
 - sobia.tariq@nu.edu.pk
- Office Hours:
 - Tuesday & Thur (11:30 AM – 12:30 PM)
- Syllabus and Grading
- Attendance 80%

Course Outline

- Intro, Agent CH 1 & 2
- Informed & Un-informed Search CH 3 & 4
- Adversarial Search CH 6
- Genetic Algorithm
- Unsupervised Learning: Clustering(K-Means & Fuzzy Clustering) CH 20
- Supervised Learning: Neural Network(SLP,MLP) and backpropagation and advance topics CH 20,21



Textbooks

1. Stuart Russell, Peter Norvig - Artificial Intelligence A Modern Approach - (3rd Edition)
2. George F. Luger - Artificial Intelligence

Reference Books

1. Tom Mitchell Machine Learning

Grading Policy



30% Midterm exam



45% Final Exam



5% Project



~10% Quizzes

Announced n unannounced quiz



**10% Programming
Assignments +
HomeWorks**

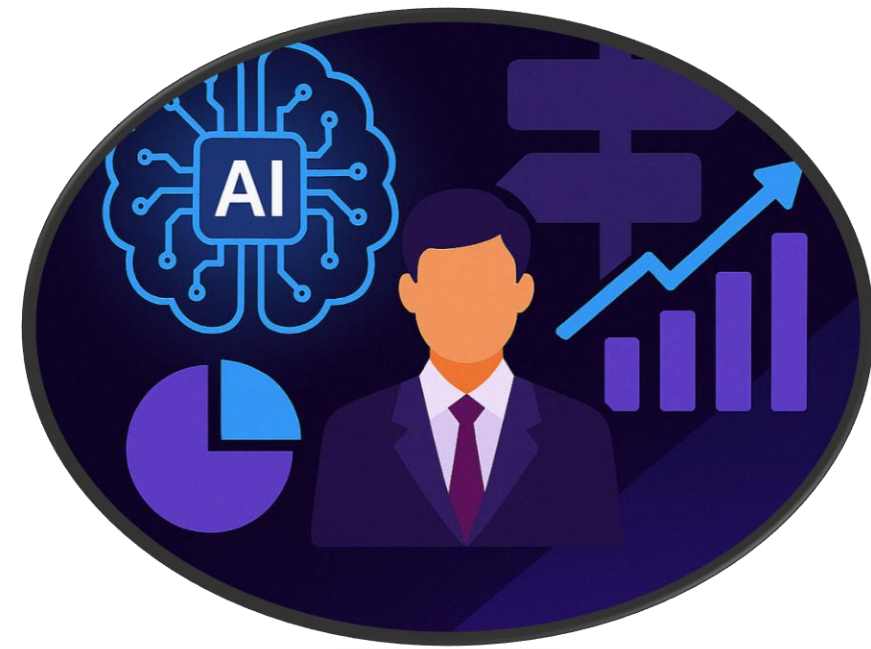
3 to 4 assignments

Publication

- S. T. Javed, K. Zafar and I. Younas "***Kids Learning Optimizer: Social Evolution and Cognitive Learning Based Optimization Algorithm***", ***Neural Computing and Applications***, <https://doi.org/10.1007/s00521-024-10009-4> vol 12, Impact Factor: 6.0 [accessed: 15th May 2024](published-2024) - **Springer**
- S. T. Javed, K. Zafar and I. Younas "***Imitation-based Cognitive Learning Optimizer for solving numerical and engineering optimization problems***", ***Cognitive Systems Research***, vol 8, 2024, <https://doi.org/10.1016/j.cogsys.2024.101237>, Impact Factor: 3.9 [accessed: 15th May 2024] (published- 2024) – **Elsevier**
- S. T. Javed, K. Zafar and I. Younas "***A decentralized metaheuristic approach to feature selection inspired by social interactions within a societal framework***", for handling datasets of diverse sizes. ***Big Data Research***, 100556 , Impact Factor: 4.2 [accessed: 15th December 2025] (published- 2025) – **Elsevier**.
- H. Mahmood, Z. Alamgir, S. T. Javed, S. Karim, and M. Awais "***Federated Generative Models in Medical Imaging: Current Advances, Challenges, and Future Directions***". ***IEEE Access***. Impact Factor: 3.7 [accessed: 15th January 2026](published- 2026) – **IEEE**.

Artificial Intelligence (AI)

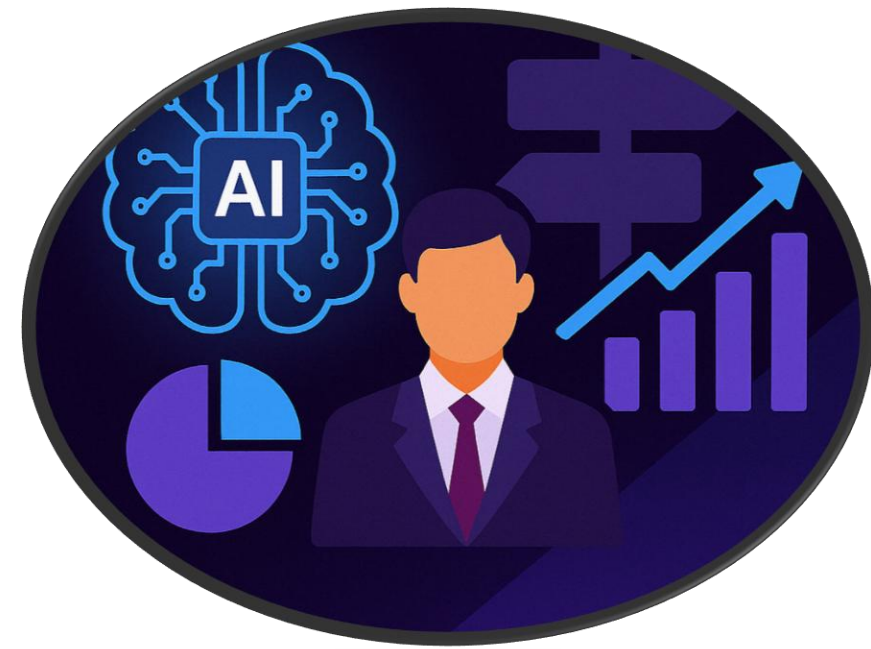
- AI is a branch of computer science that focuses on creating machines or software capable of performing tasks that normally require human intelligence.
- These tasks include:
 - Learning from data
 - Reasoning and decision-making
 - Problem solving
 - Understanding language
 - Recognizing speech and images



Artificial Intelligence (AI)

- [video](#)

AI enables computers to think, learn,
and act intelligently like humans.



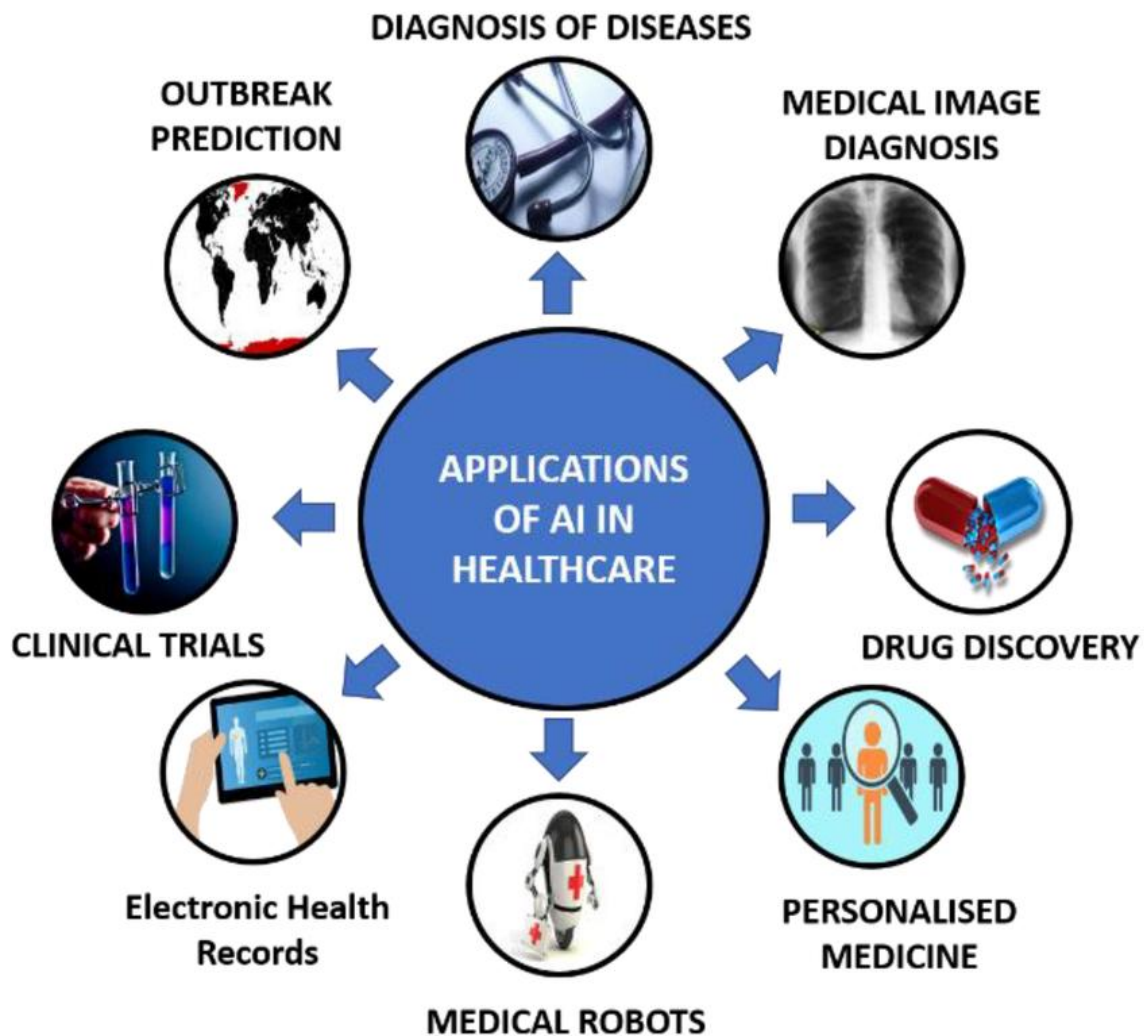
Definition of AI

- “The art of creating machines that perform functions that require intelligence when performed by people” (**Kurzweil, 1990**).
- “The branch of computer science that is concerned with the automation of intelligent behavior.” (**Luger and Stubblefield, 1993**)

Thinking Humanly “The exciting new effort to make computers think . . . <i>machines with minds</i> , in the full and literal sense.” (Haugeland, 1985) “[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning . . .” (Bellman, 1978)	Thinking Rationally “The study of mental faculties through the use of computational models.” (Charniak and McDermott, 1985) “The study of the computations that make it possible to perceive, reason, and act.” (Winston, 1992)
Acting Humanly “The art of creating machines that perform functions that require intelligence when performed by people.” (Kurzweil, 1990) “The study of how to make computers do things at which, at the moment, people are better.” (Rich and Knight, 1991)	Acting Rationally “Computational Intelligence is the study of the design of intelligent agents .” (Poole <i>et al.</i> , 1998) “AI . . . is concerned with intelligent behavior in artifacts .” (Nilsson, 1998)
Figure 1.1 Some definitions of artificial intelligence, organized into four categories.	

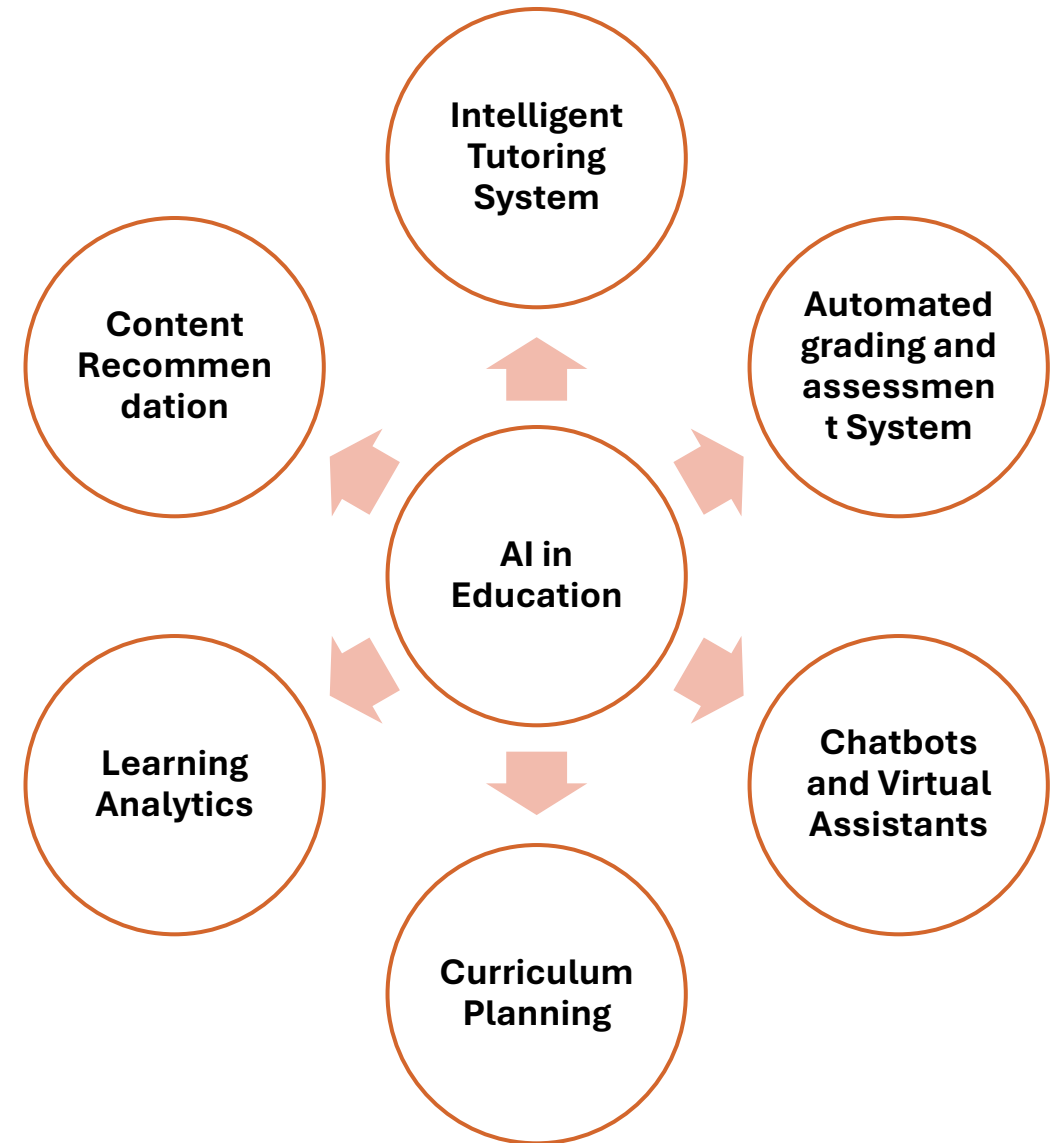
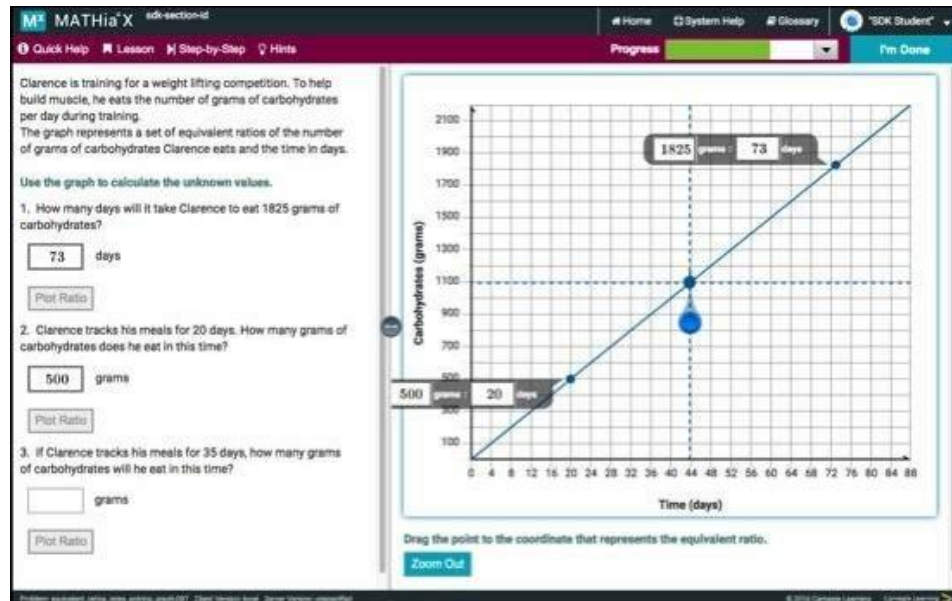
Applications of AI

- **Healthcare** – disease diagnosis, medical imaging, virtual doctors



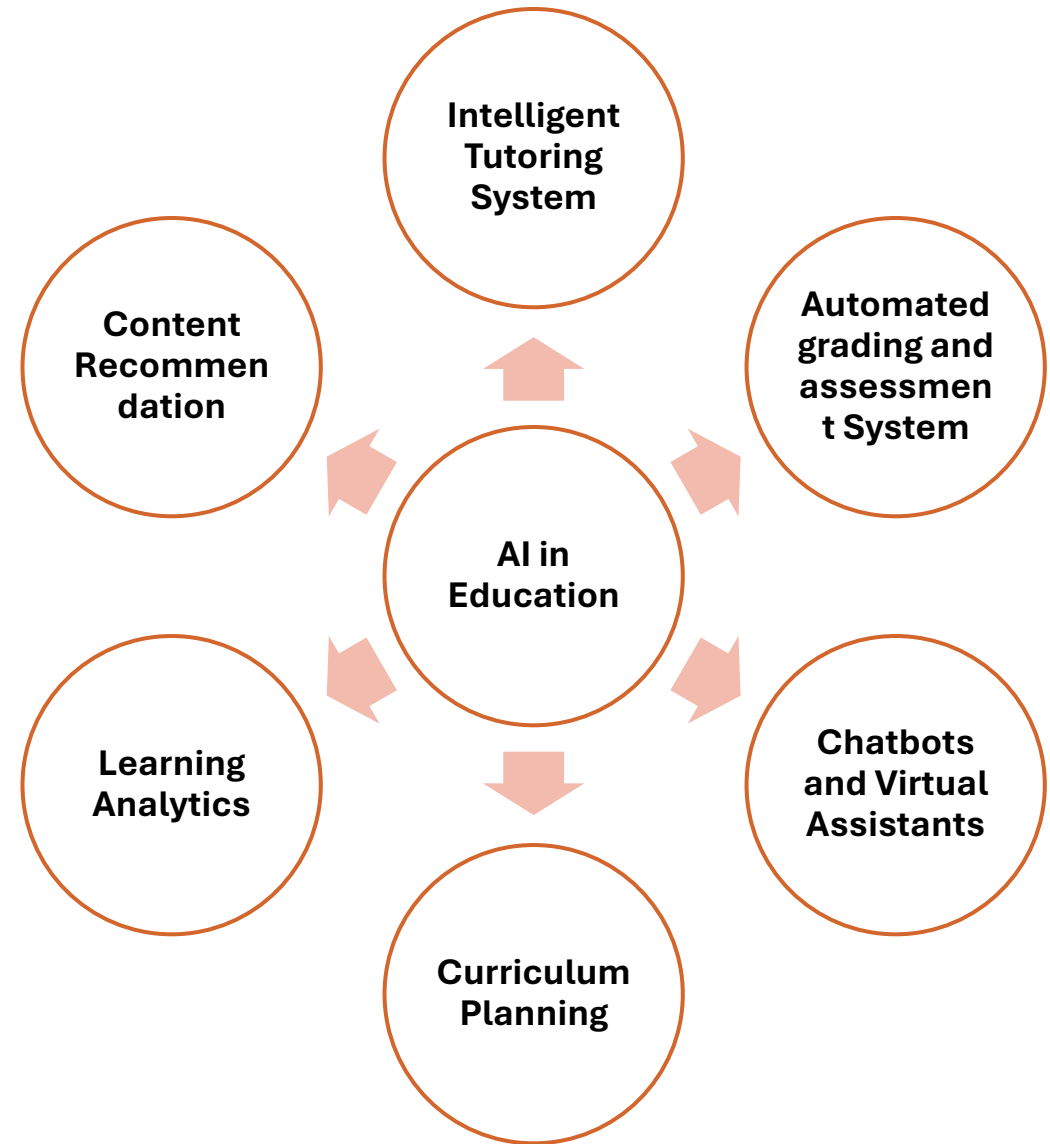
Applications of AI

- **Education** – online tutors, automated grading, personalized learning



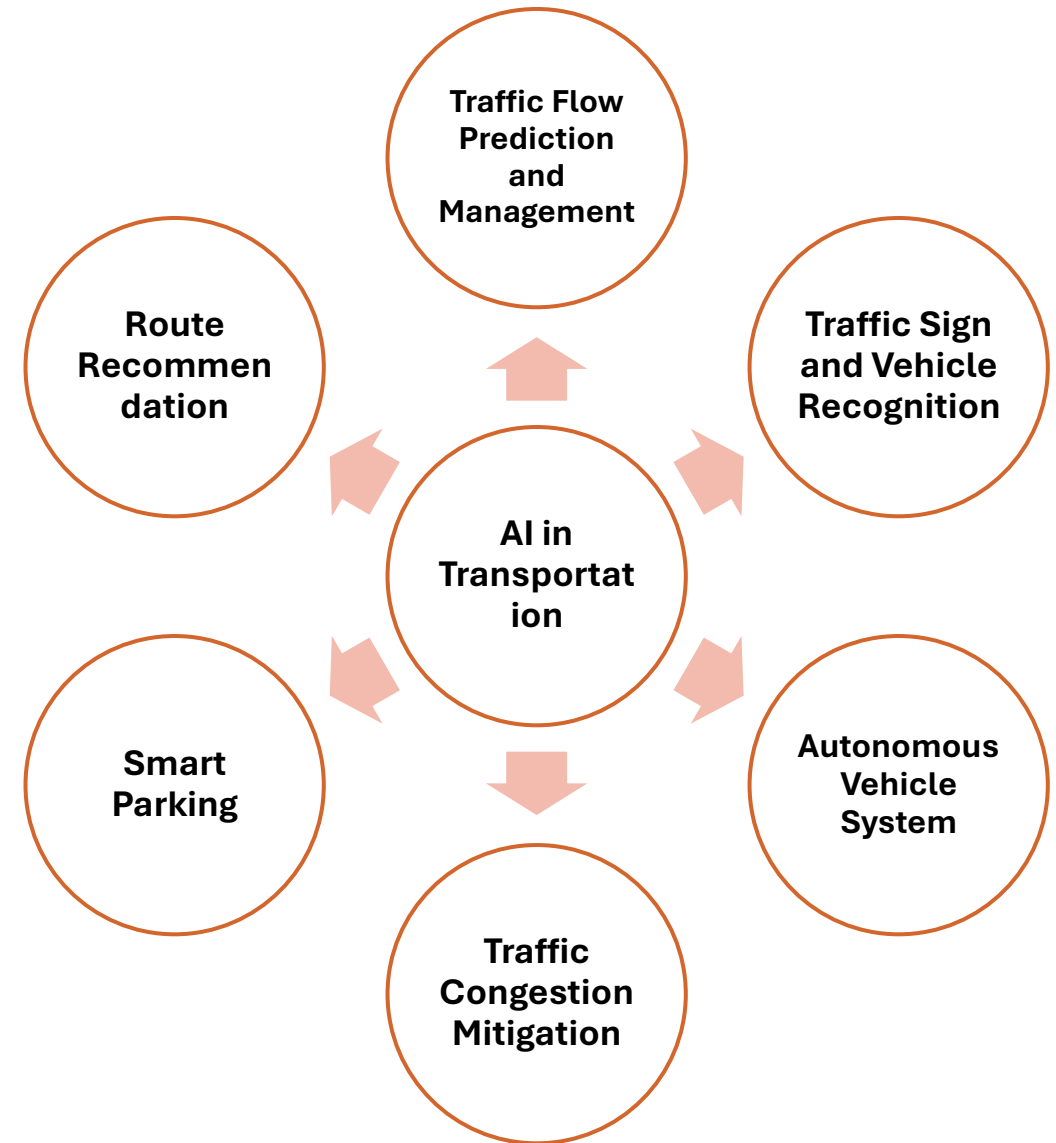
Applications of AI

- **Education** – online tutors, automated grading, personalized learning



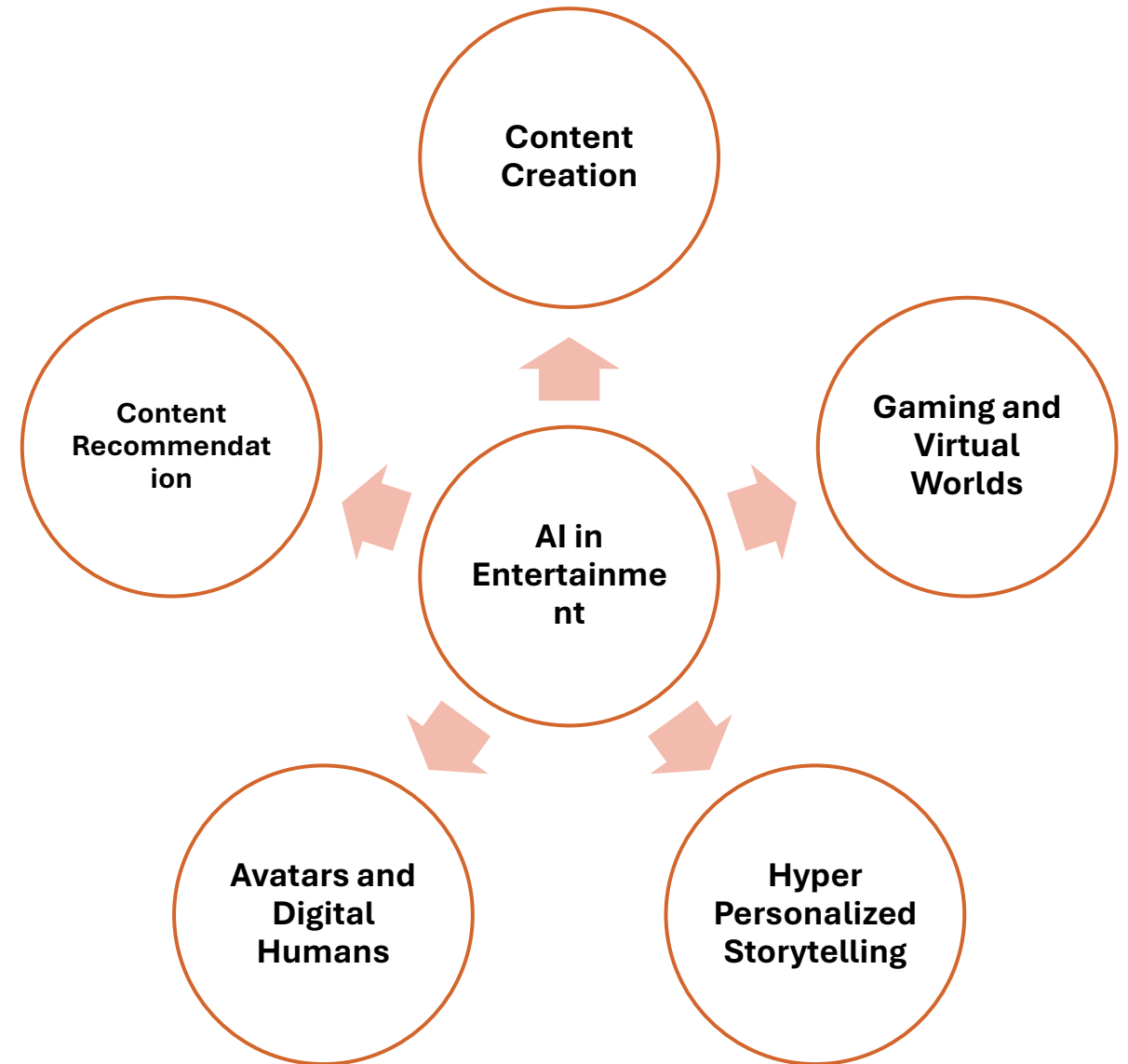
Applications of AI

- **Transportation** – self-driving cars, traffic management



Applications of AI

- **Entertainment** – game AI, movie/music recommendations

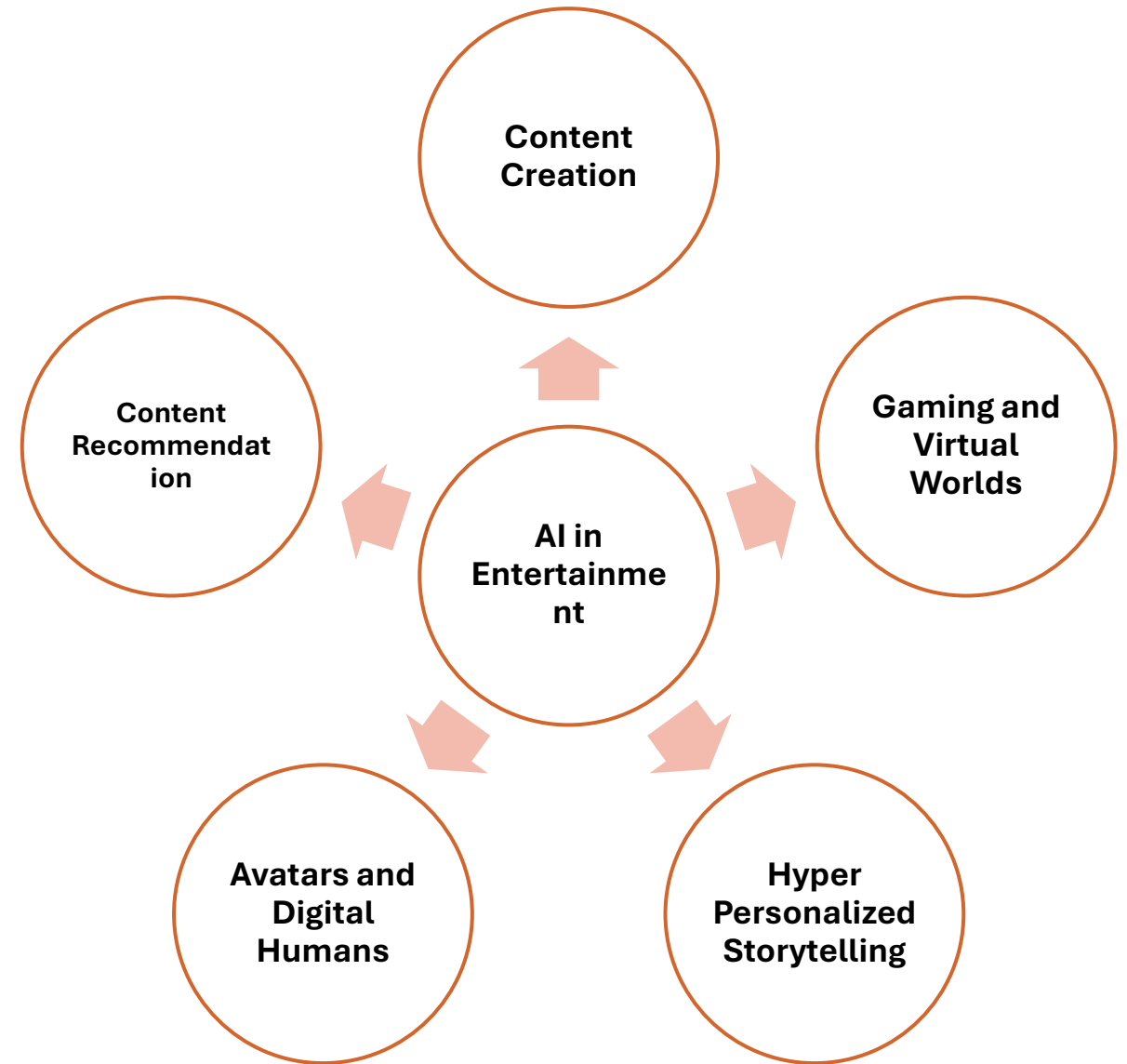




VR & Video games

Applications of AI

- **Entertainment** – game AI, movie/music recommendations



this small bird has a pink
breast and crown, and black
primaries and secondaries.



Text to Image



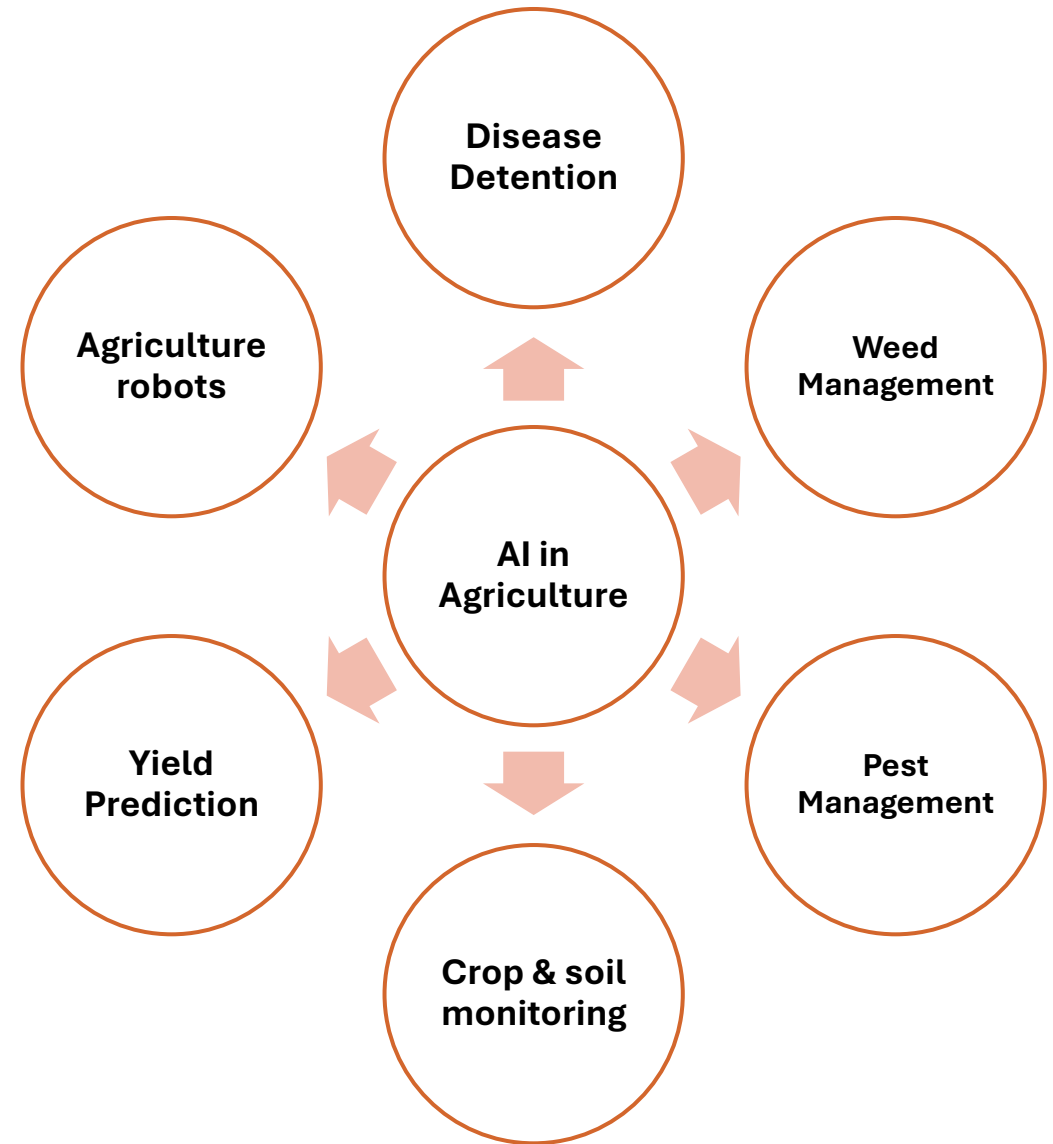
Content Creation



Avatar and digital human

Applications of AI

- **Agriculture** – crop monitoring, smart irrigation



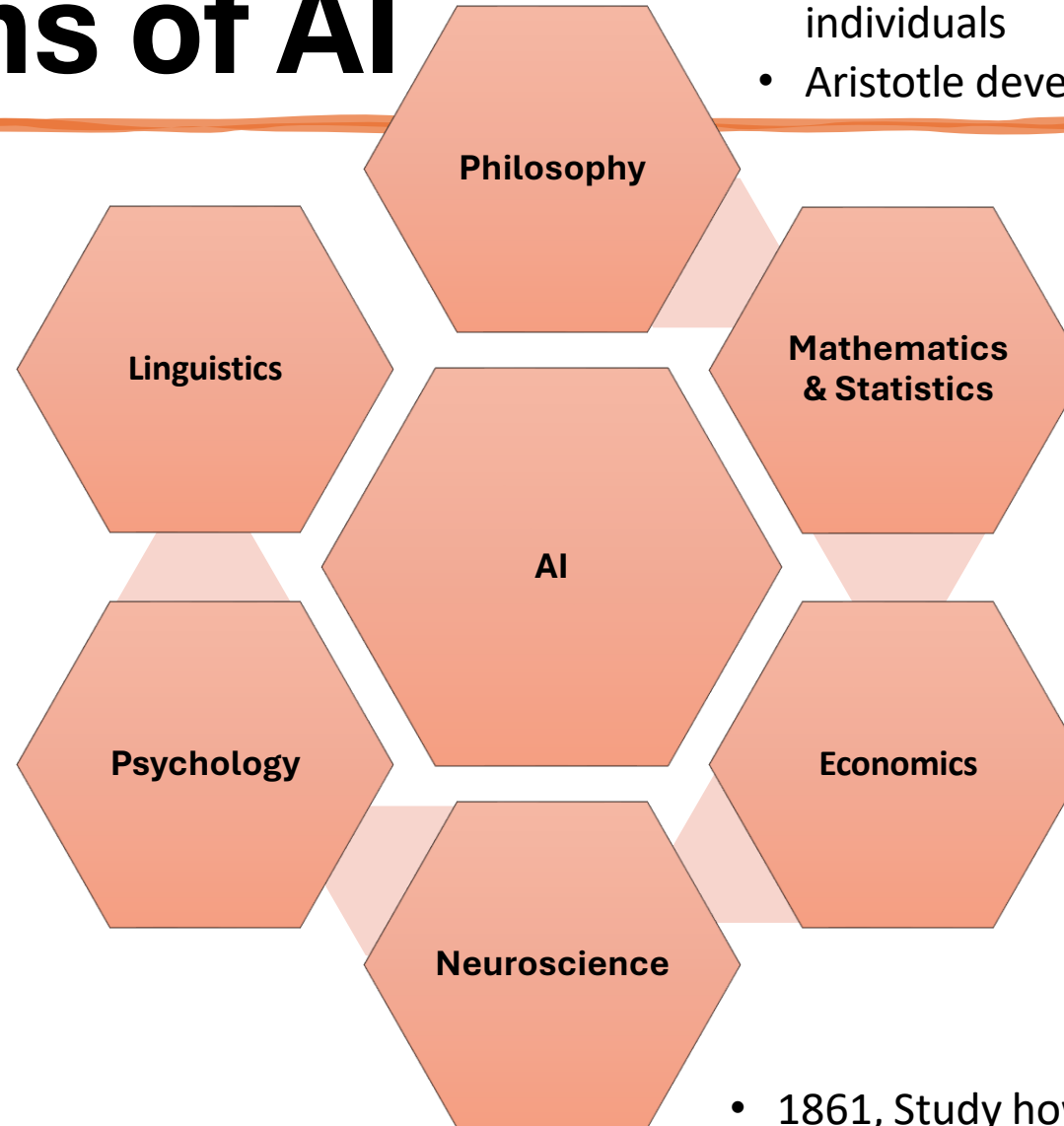
Applications of AI

- **Manufacturing** – robotics, quality control, predictive maintenance
- **Security** – face recognition, cybersecurity
- **Banking & Finance** – fraud detection, credit scoring, chatbots
- **Retail & E-commerce** – product recommendations, customer support
- **Customer Service** – chatbots and virtual assistants

Foundations of AI

- 1957, Skinner studied behaviorist approach to language learning

- 1879, Cognitive psychology initiated



- 450 BC, Socrates asked for algorithm to distinguish pious from non-pious individuals
- Aristotle developed laws for reasoning

- 1847, Boole introduced formal language for making logical inference

- 1776, Smith views economies as consisting of agents maximizing their own well being (payoff)

- 1861, Study how brains process information

Current Research

- **Gastric Cancer detection**
- **Rice Quality detection**
- **Depression detection in Children**
- **Port Flow**
- **Optimization of Auto encoders**
- **Optimization of GANs**
- **Multi-objective optimization algorithms**
- **Optimal Feature Selection**

References

- **CH 1-Introduction (Section 1.1 , 1.2)**